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# GROWTH CYCLES WITH INDUCED TECHNICAL CHANGE*

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This paper concerns itself with two related topics: the relative shares of workers and capitalists in the national income and the labour augmenting bias of technical change. Both have been dealt with, in contrasting fashion, in two seminal papers on distribution theory. Kennedy (1964) attempts to explain the then observed constancy of relative shares with the help of the now famous invention possibility frontier. However, the wage and profit rates are exogeneous to his model. Goodwin (1967), on the other hand, presents a model of cycles in growth rates in which the share of labour, the wage rate and the profit rate are endogenously determined. However, unlike the Kennedy analysis, technical change is assumed to be constant and exogeneously given.

Given that these pioneering works have addressed themselves to important questions, it seems a worthwhile task to build a model in which the shares of labour and capital, the wage rate, the profit rate, and technical change are all endogenously determined. The perspective we have chosen here introduces the key element of Kennedy's perception, induced technical change, into the Goodwin scheme and the consequences of so doing are traced out in the following terse account.

There are two classes in the economy, workers and capitalists. Only one good $q(t)$ is produced and it can be either consumed or invested. Capital stock $k(t)$ is homogeneous and, for simplicity, also assumed to be non-depreciating. The capital-output ratio at time $t$ is denoted by $\sigma(t)$. Labour, $l(t)$ is also homogeneous and its productivity at time $t$, $a(t) = q(t)/l(t)$, improves at the rate $\alpha(t)$. Labour's share in the national income denoted by $u(t)$ is

$$\frac{w(t)\,l(t)}{q(t)} = \frac{w(t)}{a(t)},$$

where $w(t)$ is the real wage. The proportion of workers employed at time $t$ is $v(t) = l(t)/n(t)$, where $n(t)$ is the total labour force at time $t$ and grows at the exogeneously given constant rate $\beta$.

The model features four theoretical elements. Firstly, there is the real wage bargaining equation which is given by:

$$\frac{\dot{w}(t)}{w(t)} = -\gamma + \rho v(t) \quad \gamma, \rho > 0. \tag{1}$$

Proportionate change in labour's share is then given by:

$$\frac{\dot{u}(t)}{u(t)} = -[\gamma + \alpha(t)] + \rho v(t). \tag{2}$$

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Secondly, capitalists invest all profits, $q(t)[1-u(t)]$, so that:

$$\frac{\dot{k}(t)}{k(t)} = \frac{[1-u(t)]}{\sigma(t)}. \tag{3}$$

Differentiating

$$v(t) = \frac{l(t)}{n(t)} = \frac{q(t)}{a(t)\,n(t)} = \frac{k(t)}{\sigma(t)\,a(t)\,n(t)}$$

then yields the proportionate change in $v$,

$$\frac{\dot{v}(t)}{v(t)} = -\frac{\dot{\sigma}(t)}{\sigma(t)} - \alpha(t) - \beta + \frac{[1-u(t)]}{\sigma(t)}. \tag{4}$$

Thirdly, there is a Kennedy–Weizsäcker technical change frontier (Samuelson, 1965) the implications of which are that the capitalists can choose either $\dot{a}(t)/a(t)$ or $-\dot{\sigma}(t)/\sigma(t)$, the choice of one fixing the other. The frontier is specified as:

$$\alpha(t) = \phi[\eta(t)], \quad \eta(t) \equiv -\frac{\dot{\sigma}(t)}{\sigma(t)}, \quad \phi' < 0, \quad \phi'' < 0. \tag{5}$$

Fourthly, it is assumed that capitalists maximise unit cost reduction,

$$\frac{\alpha wl}{q} + \eta\left(1 - \frac{wl}{q}\right) = \alpha u + \eta(1-u),$$

subject to the frontier, (5). This behaviour yields the rule:

$$\phi'(\eta) = \frac{-(1-u)}{u} \tag{6}$$

by inverting (6) we can write:

$$\eta = \psi(u), \quad \psi'(u) < 0. \tag{7}$$

Thus (5) is written as:

$$\alpha = \phi[\psi(u)]. \tag{8}$$

The system is now characterised by the following three differential equations:

$$\frac{\dot{v}}{v} = -\beta + \psi(u) - \phi[\psi(u)] + \frac{(1-u)}{\sigma}, \tag{9}$$

$$\frac{\dot{u}}{u} = -\gamma - \phi[\psi(u)] + \rho v, \tag{10}$$

$$\frac{\dot{\sigma}}{\sigma} = -\psi(u), \tag{11}$$

and the equilibrium solution is given by:

$$\sigma = \sigma^*, \quad \eta = \eta^* = 0, \quad \alpha^* = \phi(0), \quad u^* = 1 - \sigma^*(\alpha^* + \beta), \quad v^* = \frac{\gamma + \alpha^*}{\rho}. \tag{12}$$

Thus the system composed of (9)–(11) yields a steady state which is identical to that of the Goodwin model. But in this case, $\sigma$ is constant and corresponding to $\eta^* = 0$, we get the optimal $\alpha^* = \phi(\eta^*)$ which is also a constant. Therefore as long as the $\phi$ function remains constant, the Kennedy–Goodwin model collapses to the Goodwin model in the steady state.

But the introduction of induced technical progress changes the stability properties of the model. The original Goodwin model is an economic analogue of the Volterra–Lotka predator-prey model and its equilibrium is a centre around which the economy moves in closed cycles whose length is determined by initial conditions. In a sense this result is due to the implicit assumption that each side in the class struggle has only one weapon - workers can bargain on strength of full employment and capitalists can determine the growth of employment by their investment decision. Now we have given capitalists one more weapon – the choice of the induced rate of technical change. This has the effect that instead of being perpetually cyclical around the steady state, optimising behaviour leads the economy characterised by (9)–(11) to be locally stable around (12). In order to demonstrate this property let:

$$x_1 = \ln v, \quad x_2 = \ln u \quad \text{and} \quad x_3 = \ln \sigma.$$

Then (9), (10) and (11) can be rewritten as:

$$\dot{x}_1 = -\beta + \psi(e^{x_2}) - \phi[\psi(e^{x_2})] + (1 - e^{x_2})\, e^{-x_3}, \tag{9a}$$

$$\dot{x}_2 = -\gamma - \phi[\psi(e^{x_2})] + \rho e^{x_1}, \tag{10a}$$

$$\dot{x}_3 = -\psi(e^{x_2}). \tag{11a}$$

To investigate local stability, we take a linear approximation of the system around its equilibrium value and apply the Routh–Hurwitz conditions which are both necessary and sufficient (see Takayama (1974), pp. 310–1 or Murata (1977), pp. 87–93). Noting that in equilibrium $a_{11} = a_{31} = a_{23} = a_{33} = 0$, these are:

(i) $\operatorname{tr} \mathbf{A} < 0 \leftrightarrow a_{22} < 0$,

(ii) $\det \mathbf{A} < 0 \leftrightarrow a_{13} a_{21} a_{32} < 0$,

(iii) $[(a_{11} + a_{22})(a_{11} + a_{33})(a_{22} + a_{33}) - (a_{11} + a_{22})\, a_{12} a_{21} - a_{23}(a_{22} + a_{33}) - a_{23} a_{12} a_{31} - a_{13} a_{32} a_{21} - a_{13} a_{31}(a_{11} + a_{13})] < 0 \leftrightarrow (a_{22} a_{12} + a_{13} a_{32}) > 0$,

where $\mathbf{A} = (a_{ij})$, $a_{ij} = \partial \dot{x}_i / \partial x_j$. Given the assumptions of the model, conditions (i)–(iii) are easily seen to be satisfied so that the system is locally stable (for a detailed verification of these conditions and also for an investigation of global stability, see the appendix).

Our task has been to introduce a key insight of one major theory of distribution into another. The result has turned out to be quite interesting: for an arbitrarily small disturbance at the equilibrium point, the system converges back to this long run equilibrium instead of cycling around it as in the original Goodwin model. It is worth recalling that the long run equilibrium is characterised by $\eta^* = 0$ and $\alpha^* > 0$, i.e. Harrod neutral technical progress.

There are intuitively plausible reasons why the model is stable. Unlike in Goodwin's model, capitalists learn from history in this model. They invest all their profits but they also choose the new technology which will be cost minimising. Thus as long as $\phi$ remains invariant over time, any small deviation from equilibrium will allow the capitalists to claw back gains made by workers. Therefore, as long as capitalists can choose any $(\alpha, \eta)$ combination implied by the technical change frontier, the economy returns to the equilibrium state.

At the heart of this result lies the importance of allowing an extra degree of freedom to one party in a model of class struggle. In our model there is implicit a social relation whereby workers accept the capitalist's decisions on choice of techniques. But this suggests that it is in the interest of workers not to remain passive towards labour augmenting technical progress. The assumption of workers' reaction, or rather non-reaction, to technical progress now becomes a critical assumption with respect to the analysis of this paper. If workers have resisted labour augmenting bias of technical change increasingly in recent years, then the next fruitful task of theoretical research must surely be to model this in order to trace out its apparently rich consequences.

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## APPENDIX

### I. *The Verification of Conditions for Local Stability*

The Jacobian matrix **A** is:

$$\begin{bmatrix} 0 & -e^{x_2}[e^{-x_3}-\psi'(1-\phi')] & -e^{-x_3}(1-e^{x_2}) \\ \rho e^{x_1} & -e^{x_2}\phi'\psi' & 0 \\ 0 & -e^{x_2}\psi' & 0 \end{bmatrix}.$$

The necessary and sufficient conditions for local stability are:

(i) $a_{22} < 0$

$$\Rightarrow -e^{x_2}\phi'\psi' < 0.$$

Clearly satisfied as $\phi'$ and $\psi'$ are both negative.

(ii) $a_{21}a_{13}a_{32} < 0$

$$\Rightarrow (\rho e^{x_1})\{-e^{x_2}[e^{-x_3}-\psi'(1-\phi')]\}(-e^{x_2}\psi') < 0.$$

Now $\quad \rho e^{x_1} > 0, \{-e^{x_2}[e^{-x_3}-\psi'(1-\phi')]\} < 0$, and $(-e^{x_2}\psi') > 0$,

so that this condition is also satisfied.

(iii) $-(a_{22}a_{12}a_{21}+a_{13}a_{32}a_{21}) < 0$

$$\Rightarrow a_{21}(a_{22}a_{12}+a_{13}a_{32}) > 0.$$

But $a_{21} \equiv \rho e^{x_1}$ is positive. So we merely need to show that $(a_{22}a_{12}+a_{13}a_{32}) > 0$. Thus we need to show that:

$$((-e^{x_2}\phi'\psi')\{-e^{x_2}[e^{-x_3}-\psi'(1-\phi')]\}+[-e^{-x_3}(1-e^{x_2})](-e^{x2}\psi')) > 0.$$

or

$$\left(\frac{u(u-1)}{u}\psi' u\left\{\frac{1}{\sigma}-\psi'\left[1-\frac{(u-1)}{u}\right]\right\}+\frac{1}{\sigma}(1-u)u\psi'\right) > 0.$$

But this expression simplifies to $(1-u)(\psi')^2$ which is unambiguously positive. Hence this condition is also satisfied.

## II. *Investigation of Global Stability*

Our results above show only local stability of equilibrium. However, in an earlier paper Desai (1973) extended the Goodwin system by specifying a money wage Phillips curve and a pricing equation. The pricing decision gave capitalists an extra weapon and this made the extended Goodwin model *globally* stable. By analogy we should also expect global stability but this is difficult to establish analytically. *Sufficient* conditions for a system of $n$ non-linear differential equations have been established by Hartman and Olech (1962) and these require, in addition to local stability which we have already demonstrated, that:

$$\text{Max}\,[(\lambda_1+\lambda_2), (\lambda_1+\lambda_3), (\lambda_2+\lambda_3)] \leqslant 0, \tag{13}$$

where $\lambda_i$ are the eigenvalues of the matrix $\mathbf{B} = \frac{1}{2}(\mathbf{A}+\mathbf{A}')$, $\mathbf{A}$ being the Jacobian matrix previously defined in the main body of the paper. With the information at our disposal, it appears almost impossible to carry out the verification of condition (13) above. We are therefore unable to either establish or to rule out global stability of equilibrium.

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